

Mud Permeability Enables Sustained Pockmark Formation in the New England Mud Patch

Evidence for Glaciogenic Overpressure from Sensitivity-Tested Flow Simulations

Callum Hood-Cree, mentored by Dr. Brandon Dugan

Abstract

The New England Mud Patch (NEMP) hosts surface pockmarks and buried fossil pockmarks. Morphology suggests submarine groundwater discharge persists to present day, yet pockmarks are absent in the surrounding sands. Seismic evidence ~50 km west of the NEMP indicates localized overpressure ~200–300 m below seafloor. Here, we develop 2D sensitivity-tested, transient flow simulations to determine whether glaciogenic overpressure alone is physically sufficient to sustain mud-only fluidization at the NEMP up to present day across physically plausible parameter space. We find that it is, with mud permeability as the dominant control on whether and when fluidization occurs, and confining layer geometry providing a secondary control whose influence decreases over ~17 ka timescales. Results are consistent with expulsion of glaciogenic overpressured water emplaced ~17 ka persisting to present day, as recorded by the active and fossil pockmark record.

Introduction

New England Mud Patch and Pockmark Record

The NEMP is an ~13,000 km² region of active fine-grained deposition located on the continental shelf south of Cape Cod, situated at water depths between 60 m and 160 m. However, environments surrounding the mud patch are predominantly non-depositional or erosional (Goff et al., 2019). NEMP sediments, reaching a maximum thickness of about 12 m, fill a basin likely eroded by glacial outwash and overlie late-Pleistocene marine sands with distinct sand ridges (Goff et al., 2019). Sediments near the base of the unit contain a significant sand component with grain size identical to the underlying ridges, suggesting the Pleistocene sand remained exposed during the early stages of mud accumulation (Goff et al., 2019).

Pockmarks are seafloor depressions formed by upward fluid expulsion through sediment. At the NEMP, modern pockmarks up to 80 m wide and 1.2 m deep are observed on the seafloor, with reflective centers indicative of hard material and subsurface anomalies interpreted as piping features. Probable fossil pockmarks are identified as bright lenses in sub-bottom data with similar size and distribution to the modern features, suggesting episodic focused fluid flow persisting for at least 10 ka (Goff, 2019). Notably, pockmarks are confined entirely to the mud deposit and are absent from the surrounding sands. This pattern begs a physical rationale for why low-permeability muds, rather than permeable sands, display evidence of submarine groundwater discharge.

The absence of typical gas-related seismic signatures suggests submarine groundwater discharge (SGD) as the primary mechanism. These pockmarks appear morphologically fresh, supporting the characterization of the NEMP as an active modern seep environment (Goff, 2019). Approximately 50 km west of the NEMP, seismic imaging has identified velocity-inverted reflectors 200–300 m below the seafloor associated with localized overpressures of 0.9–2.2 MPa above hydrostatic levels (Siegel et al., 2014a; Goff, 2019). These pressures significantly exceed values expected from normal sediment compaction alone.

Fluid Presence and Origin

Gustafson et al. (2019) used shallow-water electromagnetic geophysical methods to image laterally continuous aquifers extending 90 km offshore New Jersey and Martha's Vineyard, indicating a continuous submarine aquifer system spanning at least 350 km of the U.S. Atlantic coast containing approximately 2,800 km³ of low-salinity groundwater. However, the origin of this water remains openly debated. One model favors glaciogenic emplacement, where late Pleistocene glaciations and subglacial recharge drove large volumes of freshwater into the continental shelf (Siegel et al., 2014b; Person et al., 2003). Alternatively, Gustafson et al. (2019) suggest meteoric recharge via onshore-connected aquifers may be feeding freshened submarine groundwater. Pockmarks observed in the NEMP provide an opportunity to test the physical plausibility of glaciogenic overpressure as a sustained fluid source, and here we determine whether glaciogenic overpressure alone is sufficient to explain the active and fossil pockmark record.

Critical Gradient

Fluid flow through porous media is governed by Darcy's Law, which relates the volumetric flux of fluid to the hydraulic gradient and the permeability of the medium (Freeze & Cherry, 1979). When upward hydraulic gradients exceed a critical threshold, the force exerted by the flowing fluid on the sediment grains overcomes the buoyant weight of the overlying material, causing fluidization, a precondition for pockmark formation. This critical hydraulic gradient is given by:

$$i_c = (G_s - 1) / (1 + e)$$

where G_s is the specific gravity of the sediment grains and e is the void ratio (Freeze & Cherry, 1979). This can be restated in terms of bulk density and fluid density:

$$i_c = (\rho_b - \rho_w) / \rho_w$$

Finer-grained sediments, such as muds, have lower permeability than sands, meaning that for a given overpressure condition, hydraulic gradients within mud are sustained for longer timescales before dissipating. This has an important implication for the NEMP: sands rapidly dissipate overpressure, but the low permeability of the overlying mud acts as a seal, retaining elevated pore pressures and sustaining gradients above the critical threshold over substantially longer timescales.

Objectives

The primary objective is to test whether an initial overpressure condition consistent with glaciogenic emplacement ~17 ka can sustain hydraulic gradients greater than the critical gradient within the NEMP mud column up to present day, while remaining below critical gradient of surrounding sands. Ultimately, this assesses whether the proposed mechanism is physically viable.

A sensitivity analysis is conducted over mud permeability, sand permeability and basal overpressure magnitude and unit thickness to determine which parameters most strongly govern whether fluidization occurs and the timing and duration of exceedance above the critical threshold.

Given uncertainty in subsurface properties at the NEMP, it is also important to quantify the impact uncertain subsurface geology may have, ensuring physically modeled results translate to reasonable geologic systems.

Methods

The subsurface is represented as a 2D vertical cross-section through the NEMP, comprising a layered stratigraphic architecture consistent with the regional geology described in Section 1. From base to top, the model includes a basal overpressured zone at depths consistent with seismic imaging by Siegel et al. (2014a), an intervening column of post-glacial shelf sediments, and an overlying mud cap. An intermediate confining layer is included in a secondary geometric sensitivity analysis to account for rapidly-deposited glacial sediments interpreted by Siegel et al. (2014a).

Transient pore pressure diffusion is governed by $S_s \partial h / \partial t = \nabla \cdot (K \nabla h)$, where S_s is the specific storage coefficient, h is hydraulic head, and K is hydraulic conductivity. The model is implemented in MODFLOW-6 using the FloPy interface, a control-volume finite-difference groundwater flow code, under the assumption of single-phase, isothermal flow with constant fluid properties ($\rho_w = 1024 \text{ kg/m}^3$). Newton-Raphson iteration with under-relaxation is used for solver convergence. The model domain is 105 km wide and extends 300 m below the seafloor, discretized into 5,000 horizontal columns ($\Delta x = 21 \text{ m}$) and 7 vertical layers. The top five layers are 2 m sublayers resolving the near-surface gradient; the remaining layers represent the bulk sub-mud sediment column and basal overpressure source. Time discretization spans ~30 ka total across four stress periods, with uniform time steps within each period. Shorter initial stress periods are included to resolve the rapid pressure diffusion through sands. Specific storage is set to $S_s = 1 \times 10^{-4} \text{ m}^{-1}$ uniform across all layers.

The basal boundary is initialized with an elevated hydraulic head consistent with glaciogenic overpressure emplaced at ~17 ka, ranging from 0.7 to 2.4 MPa above hydrostatic across the sweep. The seafloor boundary is held at hydrostatic pressure throughout the simulation. Lateral boundaries are no-flow. The upward exit gradient between the top two sublayers is evaluated at each time step and compared against the critical gradient — evaluated separately for mud and sand depending on location — and fluidization occurs when this gradient is exceeded.

Two parameter sweeps are conducted. The primary sweep varies four parameters: mud permeability (k_{mud} : 10^{-14} to 10^{-20} m^2 , with finer sampling below 10^{-18} m^2), the effective bulk vertical permeability of the sub-mud column (k_{sand} : 10^{-9} to 10^{-13} m^2), basal overpressure magnitude (0.7 to 2.4 MPa), and overpressured zone thickness (1 to 150 m). Because the subsurface architecture between the mud cap and the seismically imaged overpressure zone is unconstrained at the NEMP, k_{sand} is treated as an effective bulk vertical permeability — a heterogeneous column reduces to this single value under harmonic averaging weighted by thickness.

A secondary geometric sensitivity sweep varies mud body width, depth to the overpressure source, lateral overpressure extent, domain width, and the geometry of an intermediate confining layer, all at a fixed representative baseline drawn from the admissible region of the primary sweep. Additional scenario analyses examine the effect of intermediate seal thickness (1–100 m), the equivalence of single versus distributed seal configurations, and a laterally offset overpressure source geometry, isolating the sensitivity of fluidization outcomes to geometric assumptions not captured by the primary sweep.

Results

Across the ~6,500 simulations of the primary sweep, mud-only fluidization persists to late in the simulation record across a wide range of parameter combinations, demonstrating that glaciogenic basal overpressure is physically sufficient to sustain fluidization at the NEMP up to present day. The distribution of last-exceedance years — the time at which the upward

hydraulic gradient in the mud or sand last exceeds the critical gradient — provides the primary basis for evaluating parameter sensitivity.

Mud permeability (k_{mud}) is overwhelmingly the dominant control on fluidization timescale, with a Spearman rank correlation of $\rho = -0.96$ against last-exceedance year in the mud zone and $\rho = -0.99$ in the sand zone. Lower mud permeabilities sustain elevated gradients over longer timescales, with runs at $k_{\text{mud}} \leq 10^{-19} \text{ m}^2$ consistently producing last-exceedance years in the late Holocene or present day. At higher mud permeabilities ($k_{\text{mud}} \geq 10^{-17} \text{ m}^2$), overpressure dissipates on early post-glacial timescales and fluidization does not persist. The transition between these regimes is sharp, occurring over approximately one order of magnitude in k_{mud} around 10^{-18} – 10^{-19} m^2 (Figure 1).

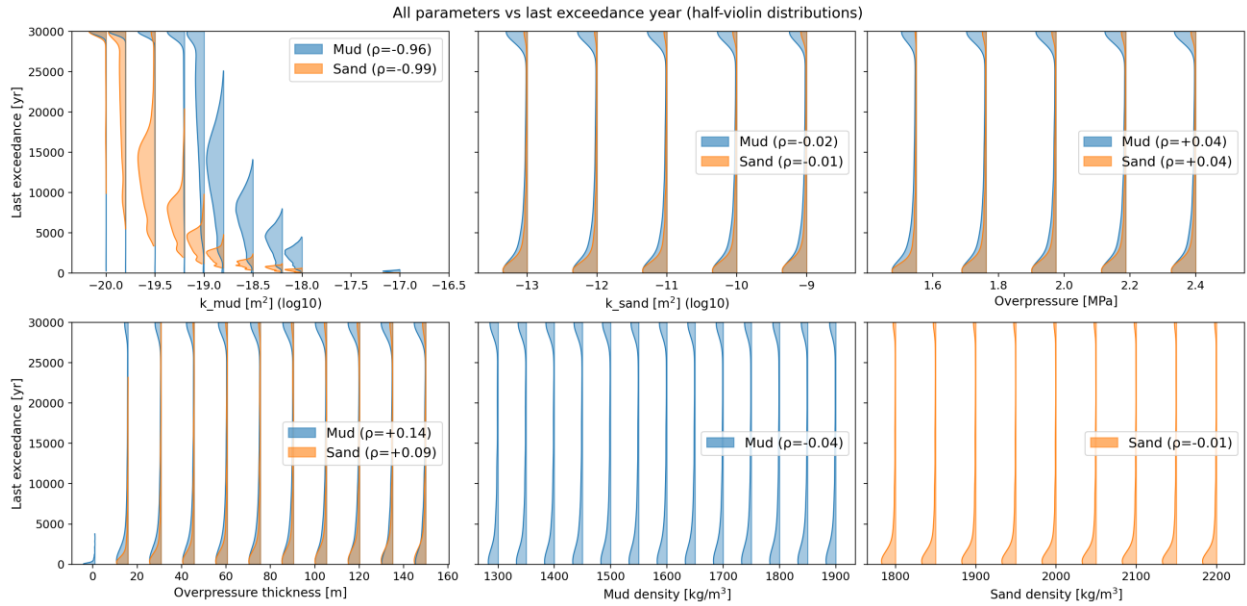


Figure 1. Half-violin distributions of last-exceedance year for mud (blue) and sand (orange) zones as a function of each swept parameter. Spearman rank correlations (ρ) are shown in legend. Mud permeability is the dominant control ($\rho = -0.96$ mud, -0.99 sand); all other parameters show negligible correlation.

In contrast, sand permeability (k_{sand}) has negligible effect on fluidization timescale in either zone ($\rho = -0.02$ and -0.01 respectively), confirming that the sub-mud sediment column does not govern when or whether fluidization occurs. This holds across four orders of magnitude in k_{sand} (10^{-9} to 10^{-13} m^2) and is consistent with the interpretation that the mud cap acts as the dominant hydraulic seal, decoupling the near-surface pressure field from the deeper sediment column. Similarly, overpressure magnitude and zone thickness show weak correlations with last-exceedance year across the full sweep ($\rho \leq +0.14$), though their influence becomes more pronounced when the analysis is restricted to runs where k_{mud} exceeds 10^{-17} m^2 . Within this high-permeability subset ($n = 96,525$), overpressure zone thickness emerges as the strongest remaining control ($\rho \approx +0.26$ for mud), followed by overpressure magnitude and k_{sand} (Figure 2).

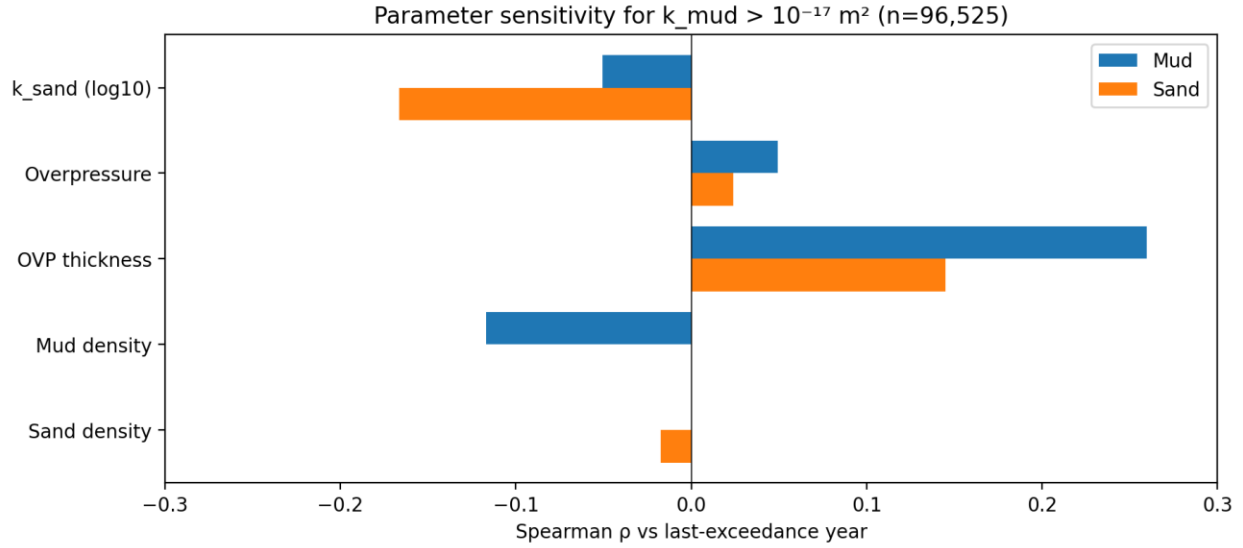


Figure 2. Spearman rank correlations between swept parameters and last-exceedance year, restricted to runs with $k_{tmu}^d > 10^{-17} \text{ m}^2$ ($n = 96,525$). Within this high-permeability subset, overpressure zone thickness is the dominant remaining control, with sand permeability and overpressure magnitude providing secondary contributions.

The geometric sensitivity sweep confirms that the primary sweep conclusions are robust to geometric uncertainty. Depth to the overpressure source has minimal impact on fluidization timescale, consistent with the dominance of the mud seal over the total diffusion pathway. The lateral extent of the overpressure zone likewise has negligible effect, as gradient exceedance is concentrated directly above the overpressure source and does not propagate laterally to any significant degree. Mud body width and domain geometry produce minor variations within the bounds of primary sweep uncertainty.

The addition of an intermediate confining layer within the sub-mud sediment column affects the gradient evolution, but its influence is strongly dependent on seal thickness. Thin seals (1–5 m) produce near-identical responses to the no-seal baseline. Thicker seals (25–100 m) at low permeability ($\sim 10^{-19} \text{ m}^2$) progressively flatten and broaden the gradient profile, sustaining elevated gradients over longer timescales in both the mud and sand zones (Figure 3). The effect is most pronounced in the first ~5,000 years and diminishes thereafter, such that the main sweep conclusions are robust at timescales consistent with the glaciogenic emplacement hypothesis.

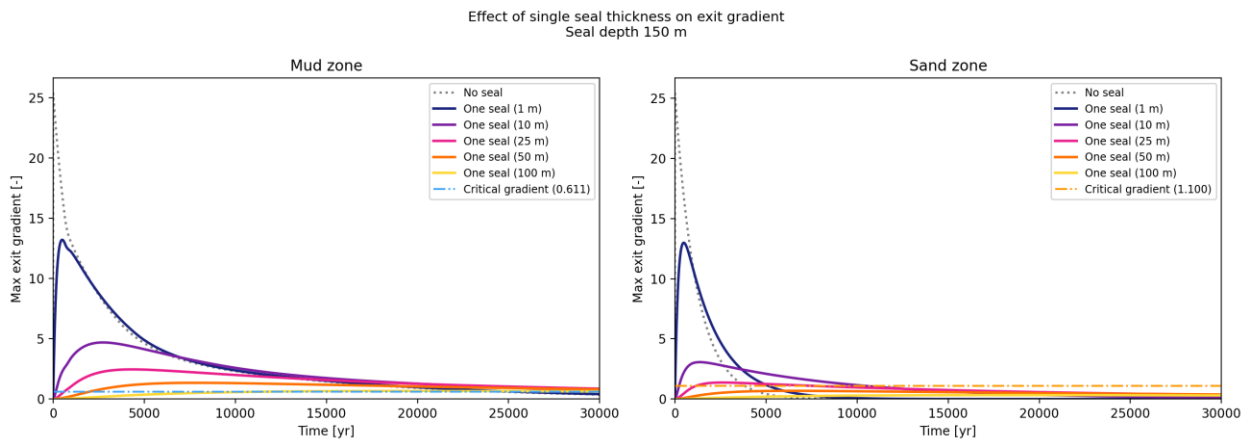


Figure 3. Maximum exit gradient in the mud zone (left) and sand zone (right) as a function of time for five seal thicknesses (1–100 m) and a no-seal baseline, at fixed baseline parameters. Critical gradients for mud (0.611) and sand (1.100) are shown as dashed lines. Thicker seals sustain elevated gradients over longer timescales; the no-seal and thin-seal (1 m) runs are near-identical.

The two-seal comparison demonstrates that a single seal of total thickness T is approximately equivalent to two seals each of thickness $T/2$ separated by a sand interlayer — the gradient profiles are near-indistinguishable (Figure 4). This confirms that the cumulative seal thickness, rather than the configuration of individual layers, governs the pressure retention effect.

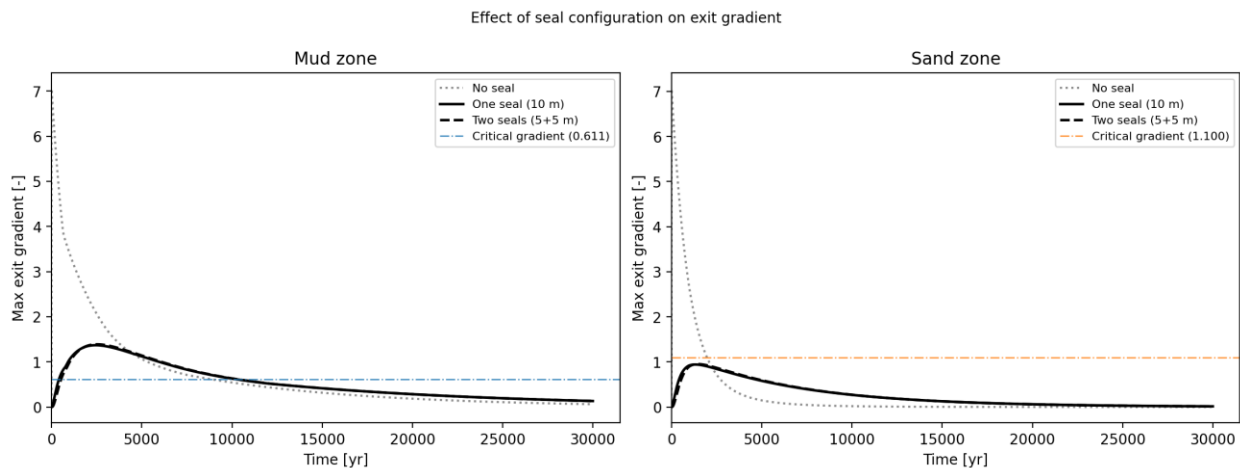


Figure 4. Comparison of exit gradient evolution for no-seal, one-seal (10 m), and two-seal (5+5 m) configurations in the mud zone (left) and sand zone (right). The one-seal and two-seal responses are near-identical, confirming that cumulative seal thickness governs pressure retention rather than the number or configuration of seal layers.

Discussion

The primary result of this study is that residual glaciogenic basal overpressure is physically sufficient to sustain mud-only fluidization at the NEMP up to present day across a wide and physically plausible range of parameter combinations. This directly addresses the knowledge gap identified in Section 1: while prior work has documented the seismic signature of overpressured fluids and morphological evidence for active SGD, no quantitative test of whether these two observations are physically consistent had been performed. The results presented here confirm that they are.

The dominance of mud permeability is the most physically meaningful finding of the sensitivity analysis. The mud cap acts as a hydraulic seal that governs the rate at which basal overpressure dissipates toward the seafloor: low-permeability muds retain elevated pore pressures over glacial timescales, while higher-permeability muds allow rapid dissipation. This explains why pockmarks at the NEMP are confined to the mud deposit rather than the surrounding sands. It is not that overpressure is concentrated beneath the mud, but that the mud retains it — sand, with its orders-of-magnitude higher permeability, dissipates the same initial overpressure too rapidly to sustain near-surface fluidization on late-Holocene timescales. The confinement of pockmarks to the mud is therefore a direct consequence of the permeability contrast between the two facies, consistent with the interpretation of Goff (2019) that pockmark concentration in the NEMP reflects the preservation potential of muddy sediments rather than a concentration of groundwater discharge.

The negligible effect of sand permeability is a clean and important secondary result. Across four orders of magnitude, $k_{\text{ta}}^{\text{nd}}$ has no meaningful influence on whether or when fluidization occurs in the mud zone. This confirms that uncertainty in the sub-mud sediment column — which is poorly constrained at the NEMP — does not undermine the conclusions, and adds credibility to the mud permeability finding.

The weak but non-zero influence of overpressure magnitude and zone thickness in the high-permeability subset indicates these parameters become relevant only when the mud seal is insufficiently restrictive. In the physically realistic low-permeability regime, the source conditions are largely irrelevant because the mud seal controls the system response. The specific details of the basal overpressure source — its magnitude, lateral extent, and depth — are therefore secondary considerations within the range consistent with the seismic observations of Siegel et al. (2014a).

The intermediate confining layer results suggest that glaciogenic interbeds, if present, would modestly extend the duration of fluidization but are not required to explain present-day pockmark activity. The two-seal equivalence result further simplifies this picture: regardless of whether glaciogenic sedimentation produced one or several low-permeability layers, the cumulative thickness governs the response and a single-layer parameterization is adequate.

These results are consistent with the glaciogenic emplacement hypothesis of Siegel et al. (2014b) and Person et al. (2003), and provide the first quantitative demonstration that this hypothesis is physically viable at the NEMP on the timescales required by the pockmark record. The results do not rule out a contribution from meteoric recharge as proposed by Gustafson et al. (2019); however, they demonstrate that glaciogenic overpressure alone is sufficient, and that no additional fluid source is required to explain the observations.

Conclusions

Residual basal overpressure consistent with glaciogenic emplacement ~17 ka is physically sufficient to sustain mud-only fluidization at the NEMP up to present day across physically plausible parameter space. Mud permeability is the dominant control on whether and when fluidization occurs. Low-permeability mud acts as a hydraulic seal that retains overpressure over glacial timescales, while permeable sand dissipates the same initial conditions too rapidly to sustain near-surface fluidization. This provides a physical explanation for the confinement of NEMP pockmarks to the mud deposit and their absence in surrounding sands.

Sand permeability has negligible influence on fluidization timescale across four orders of magnitude, confirming that uncertainty in the sub-mud sediment column does not affect the main conclusions. Overpressure magnitude and zone thickness provide secondary controls that become relevant only when mud permeability is sufficiently high to reduce seal effectiveness. Confining layer geometry provides an additional control that is most pronounced in the first ~5 ka and diminishes thereafter; cumulative seal thickness rather than layer configuration governs this effect.

Results are consistent with glaciogenic emplacement of overpressured freshwater ~17 ka beneath the NEMP persisting to present day, as recorded by the active and fossil pockmark record. No additional fluid source is required to explain the observations within the parameter space explored.

Future Work

The present study establishes physical sufficiency of glaciogenic overpressure as a fluidization mechanism at the NEMP, but several important extensions remain. A joint sensitivity sweep combining mud permeability with geometric parameters — particularly confining layer thickness and lateral overpressure extent — would determine which of these controls dominates when both are varied simultaneously. Incorporation of stratigraphic constraints from nearby seismic surveys would reduce geometric uncertainty and improve the realism of the model architecture.

The current model uses the heave gradient as a permissive screen for fluidization and does not simulate the dynamics of pipe formation, pockmark morphology, or the effect of fluid expulsion on the local pressure field. Explicit geomechanical modeling of piping and pockmark formation would allow direct comparison with the observed pockmark size and distribution reported by Goff (2019), and would provide a more physically complete test of the glaciogenic hypothesis.

The model imposes a fixed initial overpressure and does not incorporate dynamic forcing. Future work could include glacial loading as a time-varying basal pressure boundary condition and assess the contribution of meteoric recharge through shore-connected aquifers as an additional or competing fluid source. Extension to 3D geometry would capture lateral drainage pathways not represented in the 2D plane-flow assumption, which currently overestimates pressure retention relative to a fully 3D system.

Acknowledgements

Thank you to Dr. Brandon Dugan for his advisement on this project throughout the Mines Undergraduate Research Fellowship and senior design project. I also thank Dr. Ilya Tsvankin for his guidance and encouragement throughout the senior design course. Lastly, I thank all my professors and classmates for their support and encouragement throughout my time in the Mines Geophysics Department.

Data and Code Availability

Model code, input files, and sweep outputs are publicly available at https://github.com/callumhc/Pockmark_Groundwater_model.

Portions of the code development and manuscript preparation were assisted by AI language models (Claude Opus 4 and Claude Sonnet 4.6, Anthropic). All scientific content, model design, parameter choices, interpretation of results, and conclusions are the work of the author. AI tools were used under the author's direction for code structuring and document drafting assistance only.

References

- Freeze, R. A., & Cherry, J. A. (1979). *Groundwater*. Prentice-Hall, Englewood Cliffs, NJ.
- Goff, J. A. (2019). Modern and fossil pockmarks in the New England Mud Patch: Implications for submarine groundwater discharge on the middle shelf. *Geophysical Research Letters*, 46, 12213–12220. <https://doi.org/10.1029/2019GL084881>
- Goff, J. A., Reed, A. H., Gawarkiewicz, G., Wilson, P. S., & Knobles, D. P. (2019). Stratigraphic analysis of a sediment pond within the New England Mud Patch: New constraints from high-resolution chirp acoustic reflection data. *Marine Geology*, 412, 81–94. <https://doi.org/10.1016/j.margeo.2019.03.010>

Gustafson, C., Key, K., & Evans, R. L. (2019). Aquifer systems extending far offshore on the U.S. Atlantic margin. *Scientific Reports*, 9, 8709. <https://doi.org/10.1038/s41598-019-44611-7>

Person, M., Dugan, B., Swenson, J. B., Urbano, L., Stott, C., Taylor, J., & Willett, M. (2003). Pleistocene hydrogeology of the Atlantic continental shelf, New England. *Geological Society of America Bulletin*, 115(11), 1324–1343. <https://doi.org/10.1130/B25285.1>

Siegel, J., Lizarralde, D., Dugan, B., Person, M., DeFoor, W., Gable, C., & Miller, N. (2014a). Glacially generated overpressure on the New England continental shelf: Integration of full-waveform inversion and overpressure modeling. *Journal of Geophysical Research: Solid Earth*, 119, 3393–3409. <https://doi.org/10.1002/2013JB010278>

Siegel, J., Person, M., Dugan, B., Cohen, D., Lizarralde, D., & Gable, C. (2014b). Influence of late Pleistocene glaciations on the hydrogeology of the continental shelf offshore Massachusetts, USA. *Geochemistry, Geophysics, Geosystems*, 15, 4651–4670. <https://doi.org/10.1002/2014GC005569>